

Entregable Técnico: Wireshark avanzado

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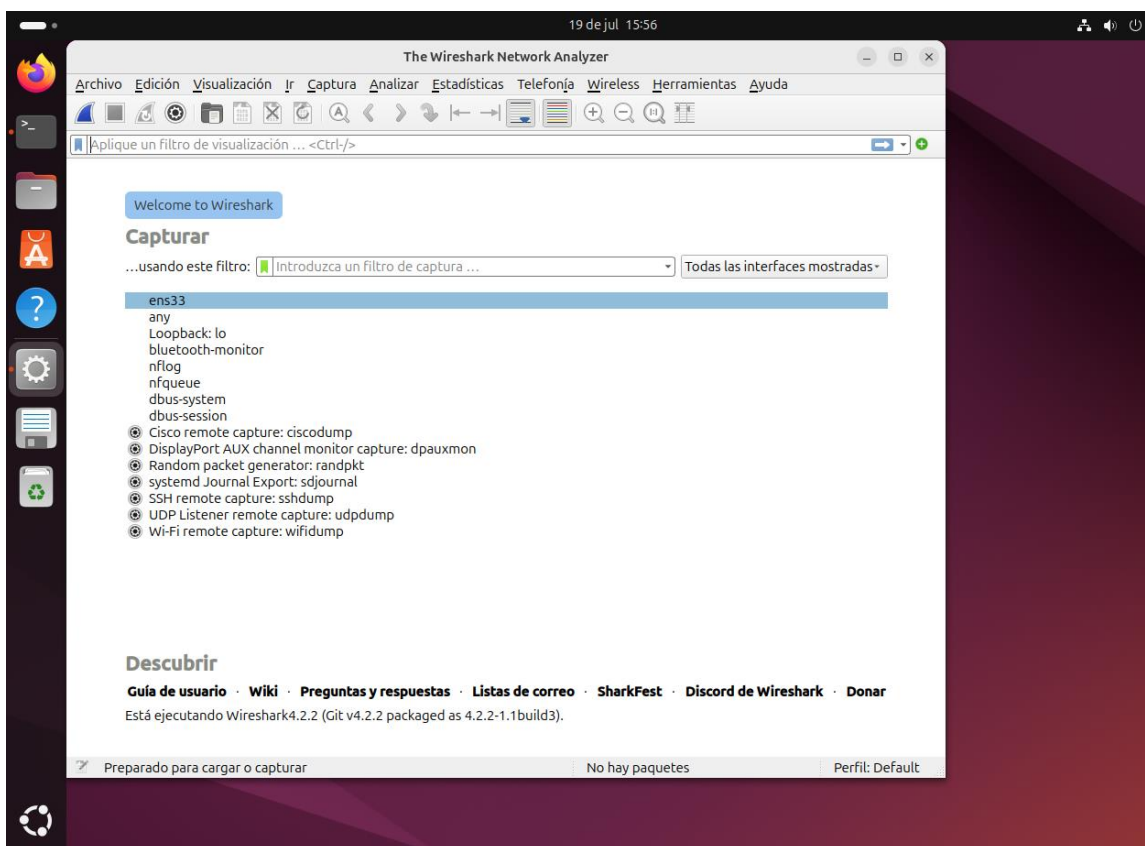
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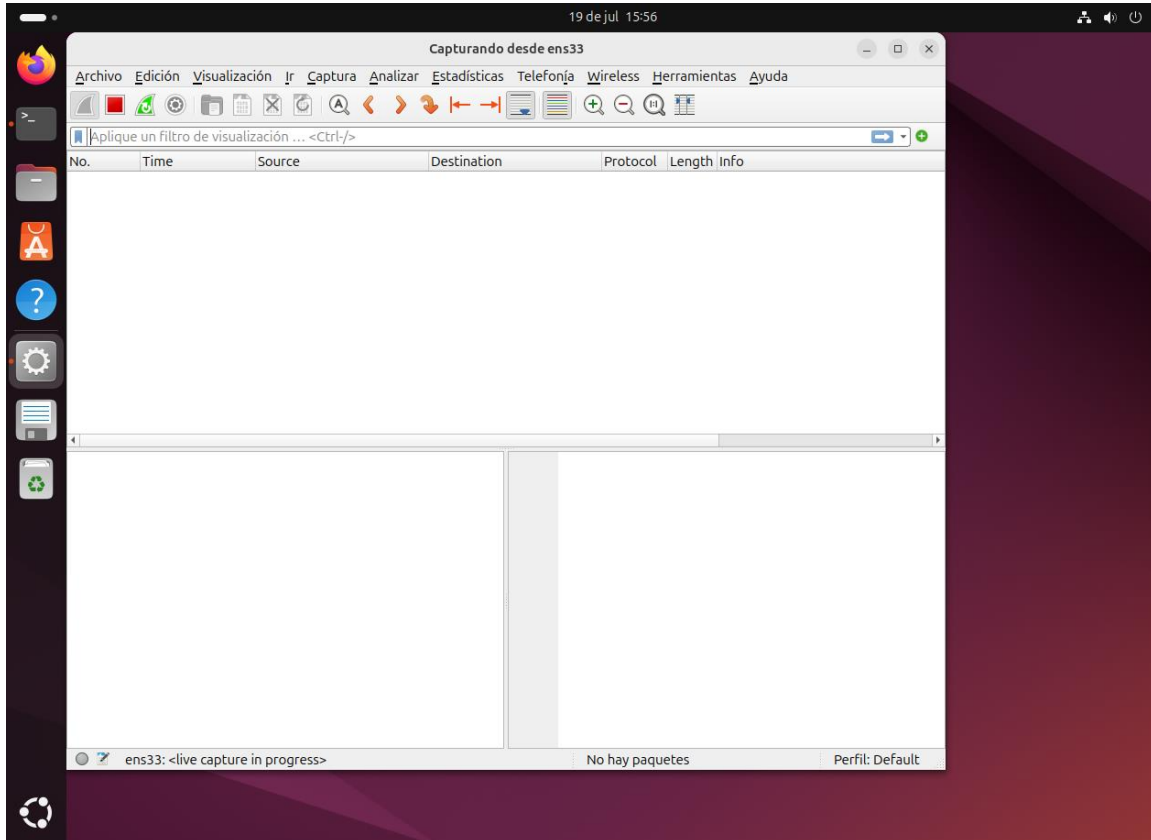
1. Objetivo del entregable y materiales necesarios

En esta práctica se utilizará Wireshark para capturar y analizar distintos tipos de tráfico de red. El objetivo es identificar protocolos habituales como DNS, HTTP, HTTPS e ICMP, aprender a utilizar filtros de visualización y comprobar qué información puede leerse en claro y cuál viaja cifrada.

2. Distintos tráficos

En primer lugar abrí Wireshark y seleccioné la interfaz de red que estaba utilizando para acceder a Internet. Una vez seleccionada, inicié la captura de paquetes para registrar todo el tráfico generado por el equipo durante la realización de la práctica.





Una vez iniciada la captura, abrí el navegador y accedí a una página web. Antes de establecer la conexión con el servidor del navegador y con otras funcionalidades fue necesario realizar consultas DNS para resolver el nombre del dominio a su dirección IP. Al aplicar el filtro `dns` pude observar tanto la consulta enviada por mi equipo como la respuesta proporcionada por el servidor DNS.

The screenshot shows a Wireshark capture of network traffic on interface *ens33. The filter is set to 'dns'. The packet list displays a series of DNS queries and responses. The packet details pane shows the structure of a DNS query packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.63.30	8.8.8.8	DNS	98	Standard query 0x1cb2 HTTPS services.addons.mozilla.org
2	0.003657642	192.168.63.30	8.8.8.8	DNS	98	Standard query 0x823e A services.addons.mozilla.org
3	0.012961828	8.8.8.8	192.168.63.30	DNS	162	Standard query response 0x823e A services.addons.mozilla.org
5	0.019245254	8.8.8.8	192.168.63.30	DNS	179	Standard query response 0x1cb2 HTTPS services.addons.mozilla.org
24	0.655874500	192.168.63.30	8.8.8.8	DNS	106	Standard query 0x0202 A content-signature-2.cdn.mozilla.net
25	0.656850945	192.168.63.30	8.8.8.8	DNS	106	Standard query 0x0a15 HTTPS content-signature-2.cdn.mozilla.net
26	0.731645878	8.8.8.8	192.168.63.30	DNS	122	Standard query response 0x0202 A content-signature-2.cdn.mozilla.net
29	0.736823594	8.8.8.8	192.168.63.30	DNS	175	Standard query response 0x0a15 HTTPS content-signature-2.cdn.mozilla.net
70	0.977513227	192.168.63.30	8.8.8.8	DNS	95	Standard query 0x3505 A detectportal.firefox.com OPT
71	0.981120828	192.168.63.30	8.8.8.8	DNS	95	Standard query 0x5c32 HTTPS detectportal.firefox.com
72	1.061662207	8.8.8.8	192.168.63.30	DNS	234	Standard query response 0x3505 A detectportal.firefox.com
73	1.061662370	8.8.8.8	192.168.63.30	DNS	228	Standard query response 0x5c32 HTTPS detectportal.firefox.com
74	1.062131418	192.168.63.30	8.8.8.8	DNS	93	Standard query 0xd5b5 HTTPS mozilla.map.fastly.net
75	1.070082613	8.8.8.8	192.168.63.30	DNS	154	Standard query response 0xd5b5 HTTPS mozilla.map.fastly.net
84	1.898830333	192.168.63.30	8.8.8.8	DNS	95	Standard query 0xf41f A api.accounts.firefox.com OPT
85	1.899459951	192.168.63.30	8.8.8.8	DNS	95	Standard query 0x53df HTTPS api.accounts.firefox.com
86	1.908903313	8.8.8.8	192.168.63.30	DNS	195	Standard query response 0xf41f A api.accounts.firefox.com
88	1.911957590	8.8.8.8	192.168.63.30	DNS	189	Standard query response 0x53df HTTPS api.accounts.firefox.com
89	1.912106387	192.168.63.30	8.8.8.8	DNS	93	Standard query 0x4499 HTTPS mozilla.map.fastly.net
91	1.951242764	8.8.8.8	192.168.63.30	DNS	154	Standard query response 0x4499 HTTPS mozilla.map.fastly.net
100	2.278709002	192.168.63.30	8.8.8.8	DNS	86	Standard query 0xa889 A ads.mozilla.org OPT
110	2.279310173	192.168.63.30	8.8.8.8	DNS	86	Standard query 0xe2ec HTTPS ads.mozilla.org OPT

Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface
 Ethernet II, Src: VMware_ea:9a:7d (00:0c:29:ea:9a:7d), Dst: VMware_e8:8f:00 (00:0c:29:e8:8f:00)
 Internet Protocol Version 4, Src: 192.168.63.30, Dst: 8.8.8.8
 User Datagram Protocol, Src Port: 52342, Dst Port: 53
 Domain Name System (query)

Domain Name System: Protocol

Paquetes: 3351 · Mostrado: 262 (7.8%)

Perfil: Default

Al aplicar el filtro `tls` observé el establecimiento de la conexión segura entre el navegador y el servidor. Aunque es posible ver información del protocolo TLS y de la conexión, el contenido intercambiado permanece cifrado.

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*ens33

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

tls

No.	Time	Source	Destination	Protocol	Length	Info
2422	7.742453055	192.168.63.30	34.120.208.123	TLSv1.3	481	Application Data
2427	7.743365030	151.101.1.91	192.168.63.30	TLSv1.2	42550	Application Data, Application Data
2428	7.743415648	151.101.1.91	192.168.63.30	TLSv1.2	21797	Application Data, Application Data
2430	7.743708139	151.101.1.91	192.168.63.30	TLSv1.2	64293	Application Data, Application Data, Application Data
2432	7.743855301	151.101.1.91	192.168.63.30	TLSv1.2	17574	Application Data
2434	7.743924113	151.101.1.91	192.168.63.30	TLSv1.2	12088	Application Data
2436	7.744395191	151.101.1.91	192.168.63.30	TLSv1.2	42550	Application Data, Application Data, Application Data
2437	7.744437452	151.101.1.91	192.168.63.30	TLSv1.2	21302	Application Data, Application Data
2439	7.746219187	151.101.1.91	192.168.63.30	TLSv1.2	42550	Application Data, Application Data, Application Data
2441	7.746296248	151.101.1.91	192.168.63.30	TLSv1.2	21797	Application Data
2442	7.746381441	151.101.1.91	192.168.63.30	TLSv1.2	7527	Application Data
2444	7.748329576	151.101.1.91	192.168.63.30	TLSv1.2	33634	Application Data, Application Data, Application Data
2446	7.748440367	151.101.1.91	192.168.63.30	TLSv1.2	30713	Application Data, Application Data
2447	7.748565727	151.101.1.91	192.168.63.30	TLSv1.2	12839	Application Data
2449	7.749645988	151.101.1.91	192.168.63.30	TLSv1.2	42550	Application Data, Application Data, Application Data
2451	7.749694809	151.101.1.91	192.168.63.30	TLSv1.2	21797	Application Data
2452	7.749766243	151.101.1.91	192.168.63.30	TLSv1.2	8814	Application Data
2455	7.750992304	151.101.1.91	192.168.63.30	TLSv1.2	42550	Application Data, Application Data, Application Data
2457	7.751073570	151.101.1.91	192.168.63.30	TLSv1.2	21797	Application Data
2460	7.753216388	151.101.1.91	192.168.63.30	TLSv1.2	64293	Application Data, Application Data, Application Data
2462	7.753414697	151.101.1.91	192.168.63.30	TLSv1.2	12839	Application Data, Application Data
2464	7.754991029	151.101.1.91	192.168.63.30	TLSv1.2	42550	Application Data, Application Data, Application Data

Frame 3254: 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interf...
 Ethernet II, Src: VMware_e8:8f:00 (00:50:56:e8:8f:00), Dst: VMware_ea:9a:7d (00...
 Internet Protocol Version 4, Src: 151.101.1.91, Dst: 192.168.63.30
 Transmission Control Protocol, Src Port: 443, Dst Port: 35550, Seq: 6611, Ack: :
 Transport Layer Security

0000 00 0c 29 ea 9a 7d 00 50 56 e8 8f 00 08 00
 0010 00 4f f8 a4 00 00 00 06 a9 7d 97 65 01 5b
 0020 3f 1e 01 bb 8a de 08 0a 77 04 99 4c 18 c2
 0030 fa f0 fc 9f 00 00 17 03 03 00 22 8c fc 49
 0040 d9 50 d1 a0 e8 62 4e bf b9 4c 54 73 39 b8
 0050 95 ef 27 d7 19 0d ea 6d 8f 68 8b bc dd

Transport Layer Security: Protocol Paquetes: 3609 · Mostrado: 1470 (40.7%) Perfil: Default

Tras aplicar el filtro HTTP observé las peticiones realizadas por el navegador y las respuestas enviadas por el servidor, pudiendo visualizar la información intercambiada al no existir cifrado en este protocolo.

Wireshark interface showing network traffic analysis. The filter is set to 'http'. The packet list displays several HTTP GET requests. The packet details pane shows the structure of a selected HTTP GET packet.

No.	Time	Source	Destination	Protocol	Length	Info
79	1.144317769	192.168.63.30	151.101.193.91	HTTP	376	GET /canonical.html HTTP/1.1
81	1.149592815	151.101.193.91	192.168.63.30	HTTP	505	HTTP/1.1 200 OK (text/html)
1648	5.456006867	192.168.63.30	151.101.193.91	HTTP	393	GET /success.txt?ipv4 HTTP/1.1
1667	5.463559588	151.101.193.91	192.168.63.30	HTTP	423	HTTP/1.1 200 OK (text/plain)
3049	16.485109681	192.168.63.30	91.189.91.97	HTTP	142	GET / HTTP/1.1
3051	16.604356929	91.189.91.97	192.168.63.30	HTTP	133	HTTP/1.1 204 No Content
3262	61.606860065	192.168.63.30	151.101.193.91	HTTP	376	GET /canonical.html HTTP/1.1
3264	61.613159243	151.101.193.91	192.168.63.30	HTTP	512	HTTP/1.1 200 OK (text/html)
3270	61.622817687	192.168.63.30	151.101.193.91	HTTP	393	GET /success.txt?ipv4 HTTP/1.1
3272	61.627619949	151.101.193.91	192.168.63.30	HTTP	425	HTTP/1.1 200 OK (text/plain)

Frame 3270: 393 bytes on wire (3144 bits), 393 bytes captured (3144 bits) on interface 0
 Ethernet II, Src: VMware_ea:9a:7d (00:0c:29:ea:9a:7d), Dst: VMware_e8:8f:00 (00:0c:29:ea:9a:7d)
 Internet Protocol Version 4, Src: 192.168.63.30, Dst: 151.101.193.91
 Transmission Control Protocol, Src Port: 43132, Dst Port: 80, Seq: 1, Ack: 1, Len: 393
 Hypertext Transfer Protocol

Paquetes: 3471 · Mostrado: 10 (0.3%) Perfil: Default

Apliqué el filtro tcp para visualizar únicamente los paquetes pertenecientes a este protocolo de transporte. Pude comprobar que la mayoría de las conexiones establecidas por el navegador utilizan TCP, ya que ofrece una transmisión fiable de los datos.

The screenshot shows the Wireshark interface with the following components:

- Menu Bar:** Archivo, Edición, Visualización, Ir, Captura, Analizar, Estadísticas, Telefonía, Wireless, Herramientas, Ayuda.
- Toolbar:** Standard network analysis tools like capture, display filter, and zoom.
- Packet List:** A table of captured packets. The selected packet is number 3274, a TCP ACK from 151.101.193.91 to 192.168.63.30.
- Packet Details:** Shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Transport Layer Security.
- Packet Bytes:** Shows the raw hex and ASCII data of the packet.
- Status Bar:** Indicates 3677 total packets, with 3304 (89.9%) displayed. The profile is set to Default.

No.	Time	Source	Destination	Protocol	Length	Info
3263	61.607131576	151.101.193.91	192.168.63.30	TCP	60	80 → 43118 [ACK] Seq=1 Ack=323 Win=64240 Len=0
3264	61.613159243	151.101.193.91	192.168.63.30	HTTP	512	HTTP/1.1 200 OK (text/html)
3265	61.613510844	192.168.63.30	151.101.193.91	TCP	54	43118 → 80 [FIN, ACK] Seq=323 Ack=460 Win=63784 Len=0
3266	61.613921329	151.101.193.91	192.168.63.30	TCP	60	80 → 43118 [ACK] Seq=460 Ack=324 Win=64239 Len=0
3267	61.616056052	192.168.63.30	151.101.193.91	TCP	74	43132 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
3268	61.622475680	151.101.193.91	192.168.63.30	TCP	60	80 → 43132 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS
3269	61.622537737	192.168.63.30	151.101.193.91	TCP	54	43132 → 80 [ACK] Seq=1 Ack=1 Win=64240 Len=0
3270	61.622817687	192.168.63.30	151.101.193.91	HTTP	393	GET /success.txt?ip=4 HTTP/1.1
3271	61.623158577	151.101.193.91	192.168.63.30	TCP	60	80 → 43132 [ACK] Seq=1 Ack=340 Win=64240 Len=0
3272	61.627619949	151.101.193.91	192.168.63.30	HTTP	425	HTTP/1.1 200 OK (text/plain)
3273	61.627918161	192.168.63.30	151.101.193.91	TCP	54	43132 → 80 [FIN, ACK] Seq=340 Ack=373 Win=63868 Len=0
3274	61.628174659	151.101.193.91	192.168.63.30	TCP	60	80 → 43132 [ACK] Seq=373 Ack=341 Win=64239 Len=0
3275	62.593613288	192.168.63.30	34.107.243.93	TLSv1.3	93	Application Data
3276	62.593796423	192.168.63.30	151.101.193.91	TLSv1.3	93	Application Data
3277	62.593858324	192.168.63.30	151.101.1.91	TLSv1.3	93	Application Data
3278	62.593959031	34.107.243.93	192.168.63.30	TCP	60	443 → 52974 [ACK] Seq=5218 Ack=2135 Win=64240 Len=0
3279	62.593959777	151.101.193.91	192.168.63.30	TCP	60	443 → 39538 [ACK] Seq=39954 Ack=3076 Win=64240 Len=0
3280	62.593959815	151.101.1.91	192.168.63.30	TCP	60	443 → 35598 [ACK] Seq=6511 Ack=3264 Win=64240 Len=0
3281	62.605535462	151.101.1.91	192.168.63.30	TLSv1.3	93	Application Data
3282	62.605535676	151.101.193.91	192.168.63.30	TLSv1.3	93	Application Data
3283	62.605535725	34.107.243.93	192.168.63.30	TLSv1.3	93	Application Data
3284	62.605588138	192.168.63.30	151.101.1.91	TCP	54	35598 → 443 [ACK] Seq=3284 Ack=6550 Win=65535 Len=0

Después utilicé el filtro de UDP para mostrar exclusivamente los paquetes que utilizan este protocolo. Entre ellos se encontraron principalmente las consultas DNS y QUIC, que habitualmente emplean UDP debido a su menor sobrecarga y mayor rapidez.

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Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

udp

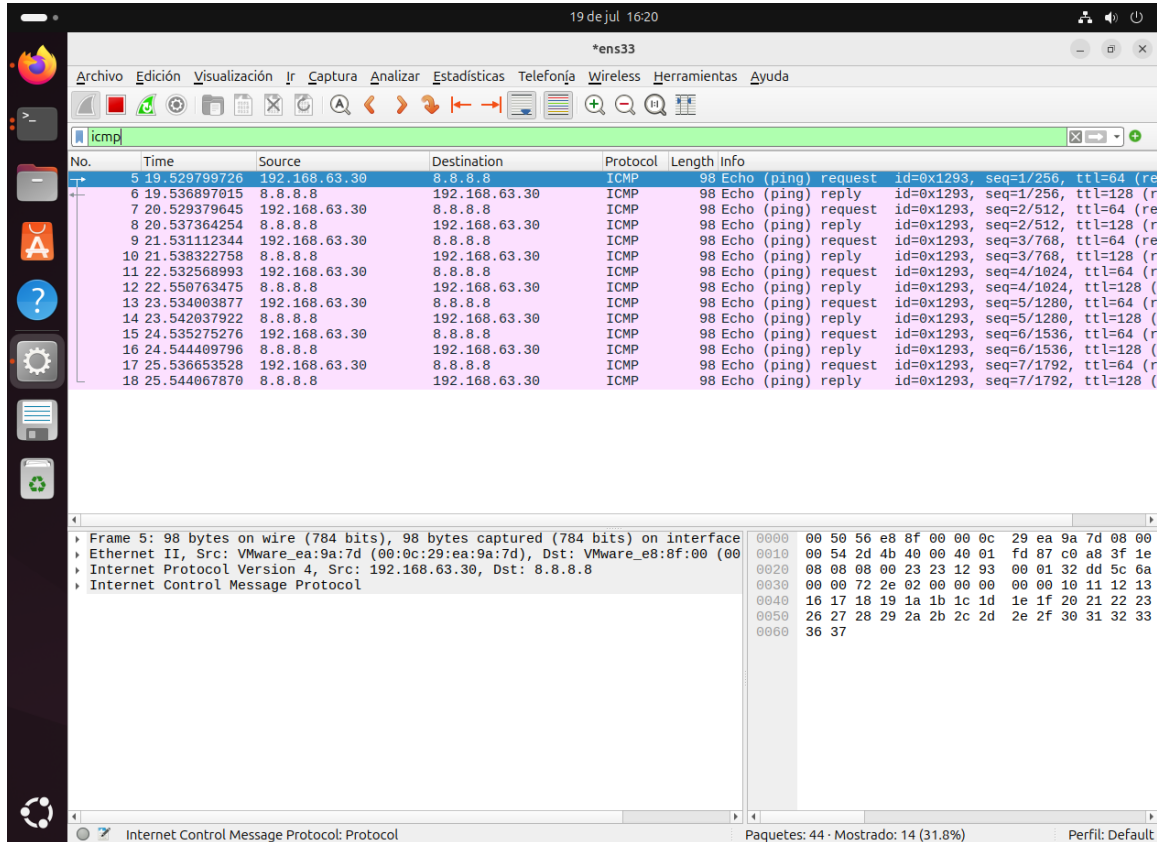
No.	Time	Source	Destination	Protocol	Length	Info
1613	5.446188053	8.8.8.8	192.168.63.30	DNS	114	Standard query response 0x7077 A example.org A 104.2
1614	5.446188149	8.8.8.8	192.168.63.30	DNS	162	Standard query response 0x259d HTTPS cloudflare-dns
1701	5.515734177	192.168.63.30	8.8.8.8	DNS	114	Standard query 0xcaad A sync-1-us-west1-g.sync.serv
1704	5.523485695	8.8.8.8	192.168.63.30	DNS	214	Standard query response 0xcaad A sync-1-us-west1-g.s
1750	5.617579329	192.168.63.30	8.8.8.8	DNS	88	Standard query 0x25c9 A relay.firefox.com OPT
1751	5.624816548	8.8.8.8	192.168.63.30	DNS	152	Standard query response 0x25c9 A relay.firefox.com A
2648	8.951123976	192.168.63.30	8.8.8.8	DNS	85	Standard query 0xfc7a HTTPS www.google.com OPT
2649	8.951887971	192.168.63.30	8.8.8.8	DNS	85	Standard query 0x2641 A www.google.com OPT
2650	8.960358610	8.8.8.8	192.168.63.30	DNS	213	Standard query response 0x2641 A www.google.com A 1
2651	8.960359230	8.8.8.8	192.168.63.30	DNS	110	Standard query response 0xfc7a HTTPS www.google.com
2652	8.967272643	192.168.63.30	142.251.153.119	QUIC	1294	Initial, DCID=0820c8d28f3e542d, SCID=4e3bac, PKN: 2
2653	8.967738947	192.168.63.30	142.251.153.119	QUIC	1294	Initial, DCID=0820c8d28f3e542d, SCID=4e3bac, PKN: 2
2654	8.979911370	142.251.153.119	192.168.63.30	QUIC	85	Initial, DCID=4e3bac, SCID=e820c8d28f3e542d, PKN: 1,
2655	8.988803258	142.251.153.119	192.168.63.30	QUIC	3798	Initial, DCID=4e3bac, SCID=e820c8d28f3e542d, PKN: 4,
2656	8.993003262	192.168.63.30	142.251.153.119	QUIC	1294	Initial, DCID=e820c8d28f3e542d, SCID=4e3bac, PKN: 2
2657	9.007691881	142.251.153.119	192.168.63.30	QUIC	4058	Protected Payload (KP0), DCID=4e3bac
2658	9.009222126	192.168.63.30	142.251.153.119	QUIC	85	Handshake, DCID=e820c8d28f3e542d, SCID=4e3bac
2659	9.013311629	192.168.63.30	142.251.153.119	QUIC	173	Protected Payload (KP0), DCID=e820c8d28f3e542d
2660	9.013632547	192.168.63.30	142.251.153.119	QUIC	116	Protected Payload (KP0), DCID=e820c8d28f3e542d
2661	9.014193389	192.168.63.30	142.251.153.119	QUIC	1294	Protected Payload (KP0), DCID=e820c8d28f3e542d
2662	9.014197657	192.168.63.30	142.251.153.119	QUIC	877	Protected Payload (KP0), DCID=e820c8d28f3e542d
2663	9.020917927	142.251.153.119	192.168.63.30	QUIC	783	Protected Payload (KP0), DCID=4e3bac

Frame 3222: 200 bytes on wire (1600 bits), 200 bytes captured (1600 bits) on in
 Ethernet II, Src: VMware_e8:8f:00 (00:50:56:e8:8f:00), Dst: VMware_ea:9a:7d (00:0c:29:ea:9a:7d)
 Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.63.30
 User Datagram Protocol, Src Port: 53, Dst Port: 45295
 Domain Name System (response)

0000 00 0c 29 ea 9a 7d 00 50 56 e8 8f 00 08 00
 0010 00 ba f8 7f 00 00 00 11 31 dd 08 08 08 08
 0020 3f 1e 00 35 b0 ef 00 a0 6f 6e bf 78 81 80
 0030 00 05 00 00 00 01 07 61 72 63 68 69 76 65
 0040 6f 7a 69 6c 6c 61 03 6f 72 67 00 00 01 00
 0050 0c 00 05 00 01 00 00 02 c9 00 22 10 6d 6f
 0060 6c 6c 61 2d 64 6f 77 6e 6c 6f 61 64 0b 66
 0070 74 6c 79 2d 65 64 67 65 03 63 6f 6d 00 c0
 0080 01 00 01 00 00 00 2d 00 04 97 65 03 13 c0
 0090 01 00 01 00 00 00 2d 00 04 97 65 83 13 c0
 00a0 01 00 01 00 00 00 2d 00 04 97 65 43 13 c0
 00b0 01 00 01 00 00 00 2d 00 04 97 65 c3 13 00
 00c0 02 00 00 00 00 00 00 00

User Datagram Protocol: Protocol Paquetes: 3680 · Mostrado: 374 (10.2%) Perfil: Default

Finalmente, realicé un ping para generar tráfico ICMP. Tras aplicar el filtro icmp observé los mensajes Echo Request enviados por mi equipo y las correspondientes respuestas Echo Reply recibidas desde el servidor remoto.



3. Conclusión y aprendizaje

Protocolos sin cifrar: DNS, HTTP, ICMP

Protocolos cifrados: HTTPS (Mediante TLS)

Mediante esta práctica he aprendido a utilizar Wireshark para capturar y analizar distintos protocolos de red. También he comprobado la utilidad de los filtros para localizar tráfico específico y he observado la diferencia entre protocolos que transmiten información en texto claro, como HTTP, y aquellos que protegen la comunicación mediante cifrado, como HTTPS.